Area-minimizing polygons with a regular difference body

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Abstract

Let $m = 3n$, and let P_m be the regular $2m$-gon of inradius one. We determine the least area of a convex polygon T satisfying $T - T = P_m$, and classify every minimizer. The minimum is $\big(m\sin(\pi/m) - \sqrt{3}\big)/\big(2\cos^2(\pi/(2m))\big)$. Equality holds precisely when the cyclic side-length sequence consists of three blocks of n maximal lengths alternating with three blocks of n zero lengths. The proof constructs a circumscribed regular hexagon from finitely many support values and derives an exact nonnegative area deficit using polygon mixed areas. The equality argument is a finite normal-cone calculation. No continuous constant-width area-minimization theorem is used.

1 Introduction and statement of the theorem

For a compact convex set $Q \subset \mathbb{R}^2$, its support function is

$$h_Q(\theta) = \max_{x\in Q} x \cdot u(\theta), \qquad u(\theta) = (\cos\theta, \sin\theta).$$

For a unit vector v, we also write $h_Q(v) = \max_{x\in Q} x \cdot v$. We use $A + B = \{a + b : a \in A,\ b \in B\}$ for Minkowski addition and $T - T = T + (-T)$ for the difference body. Area is denoted by $|T|$.

Fix an integer $n \geq 1$, and put

$$m = 3n, \qquad \delta = \frac{\pi}{m}, \qquad L = 2\tan\frac{\delta}{2}, \qquad D = \sec\frac{\delta}{2}, \qquad u_j = u(j\delta). \tag{1.1}$$

All full-grid indices are taken modulo $2m = 6n$. Let

$$P_m = \bigcap_{j=0}^{2m-1} \{x \in \mathbb{R}^2 : x \cdot u_j \leq 1\}, \qquad \mathcal{K}_m = \{T :\ T \text{ is a convex polygon and } T - T = P_m\}.$$

Thus P_m is a regular $2m$-gon of inradius one. Members of $\mathcal{K}_m$ are considered modulo translations. We retain all prescribed normal directions, assigning length zero to a collapsed side. Equiangularity refers to this support grid; the actual polygon may have fewer than $2m$ sides.

Theorem 1. *For $m = 3n$,*

$$\min_{T\in\mathcal{K}_m} |T| = A_m^* := \frac{m\sin(\pi/m) - \sqrt{3}}{2\cos^2(\pi/(2m))}. \tag{1.2}$$

Up to translation, the minimizers are exactly the grid rotations of the polygon R_m with cyclic side-length sequence

$$(L^n, 0^n, L^n, 0^n, L^n, 0^n). \tag{1.3}$$

Here L^n denotes n consecutive entries equal to L. There are $2n$ distinct minimizing sequences, forming one orbit under rotations preserving the normal grid.

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The proof is a polygonal implementation of the circumscribed-hexagon and mixed-area method associated with Chakerian [1, pp. 13–14, 17–19]. Its geometric construction and equality conditions are proved here directly for the fixed regular difference body. In particular, no Euclidean or normed-plane Blaschke–Lebesgue theorem, and no general hexagon-circumscription theorem, is assumed.

The discrete structure is explicit: the admissible class is a codimension-two slice of a cube, the auxiliary hexagon is obtained by a finite sign search, and the decisive identity is a finite weighted sum. The bound holds throughout the polytope; a reduction to its vertices is unnecessary.

2 The fixed-normal polygon model

Write $h_j = h_T(j\delta)$. The side with outward normal u_j has length

$$\ell_j = (\mathsf{D}h)_j = \frac{h_{j-1} + h_{j+1} - 2\cos\delta\, h_j}{\sin\delta}. \tag{2.1}$$

This follows by intersecting the supporting line $x \cdot u_j = h_j$ with its two neighboring supporting lines. The kernel of D consists of the two translation modes $(\cos j\delta)_j$ and $(\sin j\delta)_j$, as is also immediate from the second-order recurrence in (2.1).

Proposition 2. *Modulo translations, $\mathcal{K}_m$ is in bijection with*

$$\mathcal{P}_m = \left\{ a \in [-1,1]^m : \sum_{j=0}^{m-1} a_j u_j = 0 \right\}, \tag{2.2}$$

through the formulas

$$\ell_j = \frac{L}{2}(1 + a_j), \qquad \ell_{j+m} = \frac{L}{2}(1 - a_j), \qquad 0 \le j < m. \tag{2.3}$$

In this correspondence,

$$h_j + h_{j+m} = 1, \qquad \ell_j + \ell_{j+m} = L, \qquad \sum_{j=0}^{2m-1} \ell_j u_j = 0. \tag{2.4}$$

The correspondence is affine with respect to Minkowski addition, and $\dim \mathcal{P}_m = m - 2$.

Proof. Every facet normal of T is a facet normal of $T - T$. Indeed, exposed faces add under Minkowski addition, and a positive edge of a summand cannot disappear in the sum. Hence the normals of $T \in \mathcal{K}_m$ belong to the prescribed grid. The support identity $h_T + h_{-T} = h_{P_m}$ gives the first relation in (2.4); applying D gives the second, since $\mathsf{D}\mathbf{1} = L\mathbf{1}$. Closure of the oriented edge vectors $\ell_j J u_j$, where J is counterclockwise rotation by $\pi/2$, gives the last relation. Substitution of (2.3) yields (2.2).

Conversely, lengths (2.3) associated with $a \in \mathcal{P}_m$ are nonnegative and satisfy closure. Concatenating the vectors $\ell_j J u_j$ in angular order produces a closed convex polygon. This ordered-edge construction is elementary: starting at any prescribed edge, projection onto its outward normal first decreases and then increases to its initial value. Thus the entire chain lies on the inner side of each edge line. The chain cannot be contained in a line, since every antipodal pair of prescribed normals has positive total length. Zero lengths merely repeat a vertex.

For the resulting support vector, set $b_j = h_j + h_{j+m}$. Then $\mathsf{D}b = L\mathbf{1} = \mathsf{D}\mathbf{1}$. The vector $b - \mathbf{1}$ is antipodally even, whereas every vector in the kernel of D is antipodally odd. Thus $b = \mathbf{1}$. Between successive grid normals the support function is

$$h_T(j\delta + s) = \frac{\sin(\delta - s)h_j + \sin s\, h_{j+1}}{\sin\delta}, \qquad 0 \le s \le \delta. \tag{2.5}$$

The common supporting vertex gives this formula, including when one or more sides have collapsed. Adding the antipodal formula shows that $T - T = P_m$ in every direction.

Lengths add under Minkowski addition, and determine the polygon up to translation; this proves affinity. Finally, the closure matrix has rank two, and $a = 0$ is an interior point relative to its kernel. □

Corollary 3. *Every $T \in \mathcal{K}_m$ has the same width profile:*

$$w_T(j\delta + s) := h_T(j\delta + s) + h_T(j\delta + s + \pi) = D\cos(s - \delta/2). \tag{2.6}$$

Consequently,

$$\min_\theta w_T(\theta) = 1, \qquad \operatorname{diam} T = D, \qquad \operatorname{Per}(T) = mL, \qquad |P_m| = mL. \tag{2.7}$$

Proof. Equation (2.6) follows from (2.5) and $h_j + h_{j+m} = 1$. Its maximum is D, and the maximum width of a compact set equals its diameter. The perimeter follows by summing the paired lengths. Each side of P_m has length L; its area is half its perimeter times its inradius. □

The width in (2.6) is not constant. The condition $m = 3n$ makes it invariant under rotation through $\pi/3 = n\delta$. This sixfold symmetry is the reason that the next construction produces a regular hexagon.

3 A circumscribed hexagon from finitely many values

Define the antipodally odd part of the support function by

$$g(\theta) = \frac{h_T(\theta) - h_T(\theta + \pi)}{2}.$$

At a grid normal, $g_j = g(j\delta) = h_j - \frac{1}{2}$. Introduce

$$F(\theta) = g(\theta) + g(\theta + 2\pi/3) + g(\theta + 4\pi/3), \qquad F_j = g_j + g_{j+2n} + g_{j+4n}. \tag{3.1}$$

Since $m = 3n$ and $g_{j+m} = -g_j$,

$$F_{j+n} = -F_j. \tag{3.2}$$

The shifts in (3.1) preserve normal cells, so F obeys the same sine interpolation as (2.5).

Lemma 4. *There exist $j \in \{0, \dots, n-1\}$, $s \in [0, \delta]$, and $p \in \mathbb{R}^2$ such that the following holds. Set*

$$\phi = j\delta + s, \qquad w = D\cos(s - \delta/2), \qquad \nu_k = u(\phi + k\pi/3), \quad 0 \le k < 6. \tag{3.3}$$

Then the regular centered hexagon

$$H = \bigcap_{k=0}^{5} \{x : x \cdot \nu_k \le w/2\} \tag{3.4}$$

circumscribes $K = T - p$. In particular,

$$K, -K \subset H = -H, \qquad h_K(\nu_k) = h_{-K}(\nu_k) = w/2, \qquad |H| = \frac{\sqrt{3}}{2} w^2. \tag{3.5}$$

Proof. Inspect the finite sequence $F_0, \dots, F_n$. If one of $F_0, \dots, F_{n-1}$ vanishes, take its grid angle for ϕ. Otherwise $F_n = -F_0$ ensures that two adjacent values have opposite signs, or the latter endpoint is zero. For a strict sign change between F_j and F_{j+1}, set

$$s = \arctan\left(\frac{\sin\delta\,|F_j|}{\cos\delta\,|F_j| + |F_{j+1}|}\right). \tag{3.6}$$

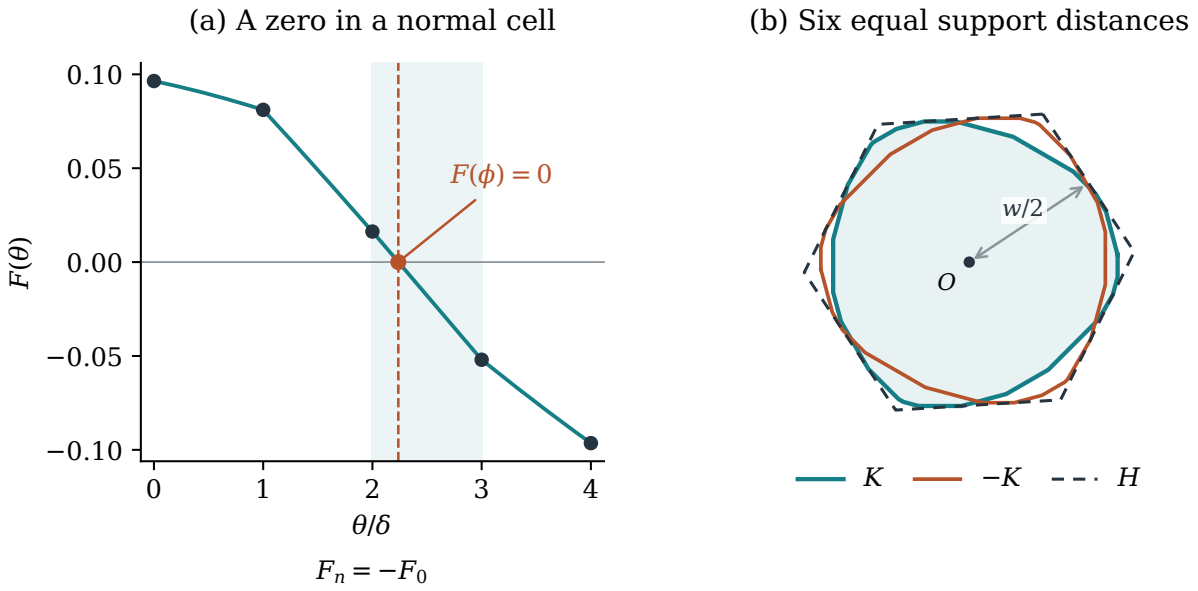


Figure 1: The finite construction for $m = 12 = 3n$. The values $F_0, \dots, F_n$ determine a zero in a highlighted normal cell by (3.6). After translating by the vector p, all six supporting distances are $w/2$. The displayed boundaries are polygons computed from finite formulas; sampling is used only to display the scalar interpolation graph.

Since $0 < \delta \le \pi/3$, this lies in $(0, \delta)$. Substitution into the sine interpolation verifies $F(j\delta + s) = 0$. Endpoint zeros can be used directly. Thus the construction requires at most n cell checks and an explicit scalar calculation.

Put $\phi = j\delta + s$. The vectors ν_0, ν_2, ν_4 sum to zero, and any two of them are independent. The equations

$$p \cdot \nu_{2k} = g(\phi + 2k\pi/3), \qquad k = 0, 1, 2,$$

are therefore compatible precisely because $F(\phi) = 0$. Solve any two equations for p. Antipodal oddness supplies the same identities for the remaining three normals.

By (2.6), the width at all six normals is the same number w: their angles differ by integer multiples of $n\delta$. Subtracting p removes the odd part of the support at those normals, leaving $h_K(\nu_k) = w/2$. All six lines in (3.4) consequently support K. Reflection gives the assertions for $-K$. The area formula is the elementary area of a regular hexagon of inradius $w/2$. □

Figure 1 illustrates the sign search and the resulting six supporting lines for an asymmetric admissible polygon.

4 Mixed areas and the discrete deficit

We record the required mixed-area identities with a finite proof. For polygons A, B, define

$$V(A, B) = \frac{1}{2} \sum_{e \subset \partial A} |e|\, h_B(\nu_e), \tag{4.1}$$

where ν_e is the outward unit normal of the edge e.

Lemma 5. *Polygon mixed area is symmetric, translation invariant, and monotone under inclusion in either argument. Moreover,*

$$V(A, A) = |A|, \qquad |A + B| = |A| + 2V(A, B) + |B|. \tag{4.2}$$

Proof. Use a common refinement of the two normal grids, inserting additional normals if necessary so that successive angular gaps α_i lie in $(0, \pi)$. For a polygon with support vector b, its edge lengths on this grid are

$$(\mathsf{M}b)_i = \frac{b_{i-1}}{\sin \alpha_{i-1}} + \frac{b_{i+1}}{\sin \alpha_i} - (\cot \alpha_{i-1} + \cot \alpha_i) b_i.$$

The matrix M is symmetric. The polygon area formula, obtained by summing signed triangles from the origin to the edges, is $|A| = \frac{1}{2} a^{\mathsf{T}} \mathsf{M} a$. Hence $V(A, B) = \frac{1}{2} b^{\mathsf{T}} \mathsf{M} a = \frac{1}{2} a^{\mathsf{T}} \mathsf{M} b = V(B, A)$. Adding support vectors and expanding the quadratic form proves (4.2). Zero lengths cause no change.

Translation invariance in the second argument follows from $\sum_e |e| \nu_e = 0$, and symmetry supplies it in the first. If $B \subset B'$, then $h_B \le h_{B'}$, so the nonnegative weights $|e|$ in (4.1) give $V(A, B) \le V(A, B')$. Symmetry again supplies the other argument. □

Let K, H, w, s be supplied by Lemma 4. Because every side of H supports K, the finite support sum gives

$$V(K, H) = V(H, K) = V(H, H) = |H|. \tag{4.3}$$

For every prescribed normal define

$$d_j = h_H(u_j) - h_{-K}(u_j) \ge 0. \tag{4.4}$$

These gaps are computed from six vertices:

$$z_k = \frac{w}{\sqrt{3}}\, u\big(\phi + \pi/6 + k\pi/3\big), \qquad h_H(u_j) = \max_{0 \le k < 6} z_k \cdot u_j. \tag{4.5}$$

Also $h_{-K}(u_j) = h_{j+m} + p \cdot u_j$.

Proposition 6 (Exact deficit formula)**.** *Every $T \in \mathcal{K}_m$ satisfies*

$$\begin{aligned} |T| - A^*_m &= \frac{\sqrt{3}}{2}(D^2 - w^2) + \frac{1}{2} \sum_{j=0}^{2m-1} \ell_j d_j \\ &= \frac{\sqrt{3}}{2} D^2 \sin^2(s - \delta/2) + \frac{1}{2} \sum_{j=0}^{2m-1} \ell_j d_j \ \ge 0. \end{aligned} \tag{4.6}$$

This identity holds for every zero chosen in the construction of Lemma 4.

Proof. Equations (4.3) and (4.1) imply

$$|H| - V(K, -K) = V(K, H) - V(K, -K) = \frac{1}{2} \sum_{j=0}^{2m-1} \ell_j d_j.$$

On the other hand $K - K = P_m$, so polarization gives

$$|P_m| = 2|T| + 2V(K, -K).$$

Eliminating the mixed area yields

$$|T| = \frac{|P_m|}{2} - |H| + \frac{1}{2} \sum_j \ell_j d_j.$$

Now use $|P_m| = mL$, $|H| = \sqrt{3}\, w^2/2$, and

$$\frac{mL}{2} - \frac{\sqrt{3} D^2}{2} = \frac{m \sin \delta - \sqrt{3}}{2 \cos^2(\delta/2)} = A^*_m.$$

Finally $w = D \cos(s - \delta/2)$, and every length and gap in the remaining finite sum is nonnegative. □

The first term in (4.6) records the angular displacement of the hexagon normals from the midpoints of the normal cells. The second records the support gap between $-K$ and H, weighted by the edge lengths of K. Thus equality controls both the position of the auxiliary hexagon and the location of every positive edge normal.

5 Attainment by three equal blocks

Consider the lengths in (1.3). Their three positive blocks are related by rotation through $2\pi/3 = 2n\delta$, so their edge-vector sums close. They also satisfy $\ell_j + \ell_{j+m} = L$. Proposition 2 therefore constructs a polygon $R_m \in \mathcal{K}_m$.

Each block is a chain of n equal edges with successive turning angles δ. Since $L = 2D\sin(\delta/2)$, the chain consists of n chords of a circle of radius D, with total central angle $n\delta = \pi/3$. Its endpoint chord has length

$$L\frac{\sin(n\delta/2)}{\sin(\delta/2)} = D. \tag{5.1}$$

The three endpoint chords consequently form an equilateral triangle of side D. For each chain, the circle center is the opposite vertex of this triangle: it is at distance D from both endpoints and lies on the inner side of the chain. The decomposition is shown in Figure 2.

The polygonal cap between one chain and its endpoint chord has area

$$\frac{D^2}{2}\left(n\sin\delta - \sin\frac{\pi}{3}\right). \tag{5.2}$$

Indeed, join the $n+1$ chain vertices to its circle center. The resulting n triangles have area $D^2\sin\delta/2$ each. Subtract the triangle joining the center to the two endpoints, of area $D^2\sin(\pi/3)/2$. This proves (5.2) by a finite triangle sum.

Adding the central equilateral triangle and the three caps gives

$$\begin{aligned} |R_m| &= \frac{\sqrt{3}D^2}{4} + \frac{3D^2}{2}\left(n\sin\delta - \frac{\sqrt{3}}{2}\right) \\ &= \frac{D^2}{2}(m\sin\delta - \sqrt{3}) = A_m^*. \end{aligned} \tag{5.3}$$

For $n = 1$, the caps have zero area and R_3 is an equilateral triangle. Proposition 6 and (5.3) establish the minimum value in Theorem 1.

6 Equality forces the three-block pattern

We finish by proving uniqueness without assuming in advance that a minimizer has saturated lengths. Figure 3 shows the contact and normal-cone configurations forced by equality.

Lemma 7. *A subset of the vertices of a regular hexagon containing no opposite pair either is an alternating triple or is contained in three consecutive vertices.*

Proof. Such a set contains at most three vertices. A set of at most two vertices with no opposite pair fits in three consecutive vertices. For a set of three, choose one vertex from each of the three opposite pairs. Of the eight possible choices, two are the alternating triples and the remaining six are precisely the triples of consecutive vertices. □

Completion of the proof of Theorem 1. Suppose $|T| = A_m^*$, and use the hexagon supplied by Lemma 4. Both nonnegative terms of (4.6) vanish. The first gives

$$s = \delta/2, \qquad w = D. \tag{6.1}$$

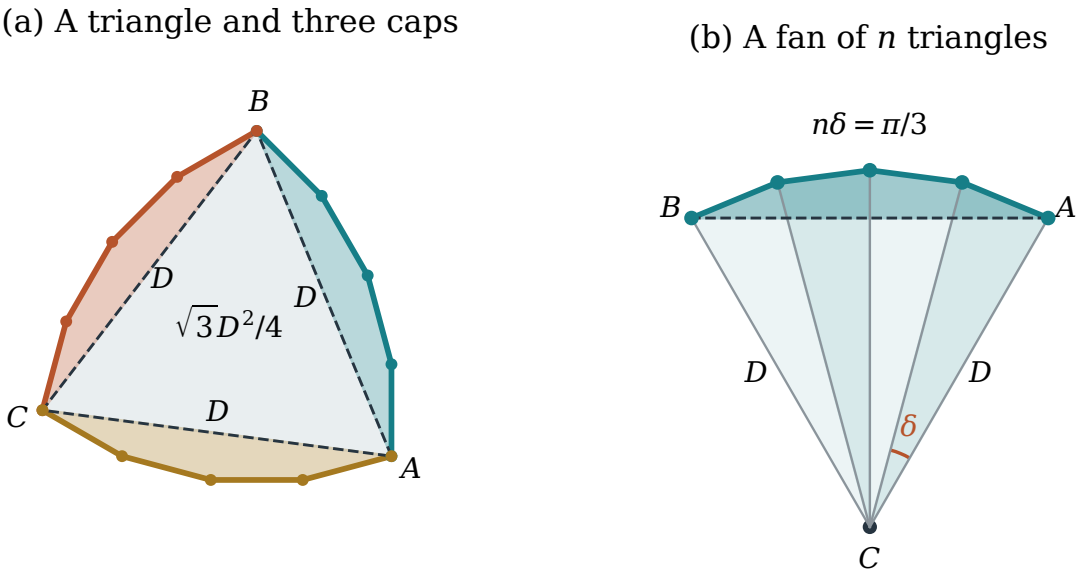


Figure 2: Sharpness for $m = 12$, $n = 4$. Three chains of n maximal edges surround an equilateral triangle of side D. For the chain from A to B, the opposite vertex C is its circle center. The cap area is the sum of n isosceles-triangle areas minus $|\triangle CAB|$. All shaded regions have straight polygonal boundaries; no circular-sector area is used.

Hence every facet normal $\nu_k = u(\phi + k\pi/3)$ of H lies halfway between two original grid normals.

For the vertex z_k in (4.5), let C_k denote its open normal cone, viewed as an arc of the unit circle:

$$C_k = \{u(\theta) : \ \phi + k\pi/3 < \theta < \phi + (k+1)\pi/3\}.$$

Angles are interpreted cyclically. Since $\pi/3 = n\delta$, each C_k contains exactly n grid normals. No grid normal lies on a cone boundary.

If $\ell_j > 0$, the vanishing of the second term in (4.6) forces $d_j = 0$. There is a unique vertex z_k of H maximizing the linear functional $x \mapsto x \cdot u_j$, because u_j lies in an open cone. As $-K \subset H$ and $h_{-K}(u_j) = h_H(u_j)$, this vertex must belong to $-K$. Define

$$I = \{z_k : \ z_k \text{ is a vertex of } H \text{ and } z_k \in -K\}.$$

Every positive edge normal of K lies in a cone associated with a vertex in I.

Opposite hexagon vertices are at distance $2D/\sqrt{3}$, while $\operatorname{diam}(-K) = D$ by Corollary 3. Thus I contains no opposite pair. By Lemma 7, either I is an alternating triple or it is contained in three consecutive vertices.

The latter possibility contradicts edge closure. Indeed, the three associated open cones lie in an open semicircle: their total angular span is π, and its outer endpoints are half-grid directions. Every normal with $\ell_j > 0$ would then satisfy $q \cdot u_j > 0$ for a suitable unit vector q. Since at least one edge length is positive, this would imply

$$0 = q \cdot \sum_{j=0}^{2m-1} \ell_j u_j = \sum_{j=0}^{2m-1} \ell_j (q \cdot u_j) > 0,$$

which is impossible.

Consequently I is an alternating triple. All lengths in the other three cones vanish. The antipodal map takes each selected cone to an unselected cone. The relation $\ell_j + \ell_{j+m} = L$ therefore forces every length in a selected cone to equal L. Each cone contains exactly n grid

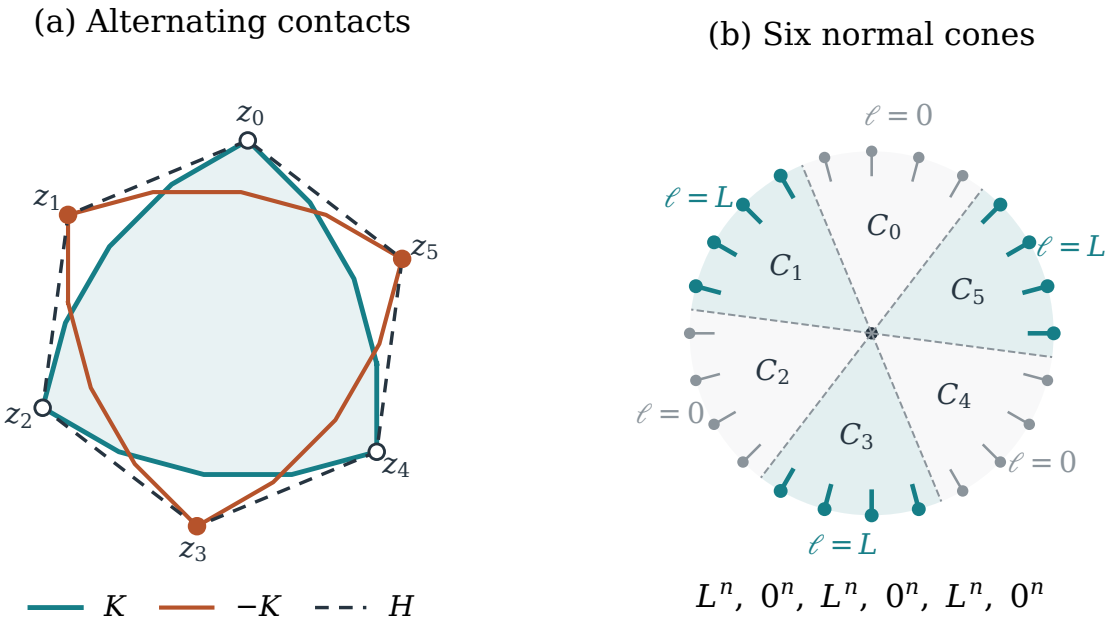


Figure 3: The equality mechanism for $m = 12$, $n = 4$. Filled rust-colored vertices of H belong to $-K$. On the circle of normal directions, the cones associated with those vertices contain the positive edge normals of K. Each cone contains n grid directions, and none lies on a boundary ray. Antipodal pairing then forces the alternating values L and 0.

normals, so the cyclic sequence is $(L^n, 0^n, L^n, 0^n, L^n, 0^n)$, up to a cyclic shift. These lengths determine T up to translation.

The sequence has least cyclic period $2n$, so its shifts produce exactly $2n$ distinct support configurations modulo translations. This proves every assertion of the theorem. □

Remark 8 (Positive side lengths and scaling). If every prescribed side is required to have positive length, the same number A_m^* is the infimum and is not attained. Indeed, equality has zero lengths, whereas the parameters $(1 - t)a(R_m)$, $0 < t < 1$, produce positive lengths and converge to the minimizing polygon as $t \downarrow 0$. If the inradius of the prescribed regular difference polygon is $b > 0$, dilation multiplies the minimum by b^2.

References

[1] G. D. Chakerian, *Sets of constant width*, Pacific Journal of Mathematics **19** (1966), no. 1, 13–21. Publisher PDF.